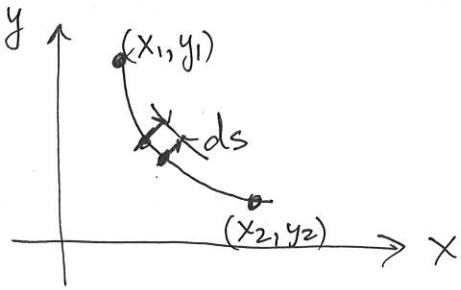
Bernoulli 1696 → Euler 1744

Brachistochrone

Fastest frictionless sliding from $P_1 = (x_1, y_1)$ to $P_2 = (x_2, y_2)$.



$$dt = \frac{ds}{v} \quad \text{where } ds = \text{length of trajectory}$$

$$ds = \sqrt{1 + \left(\frac{dy(x)}{dx}\right)^2}\, dx = \sqrt{1 + (y')^2}\, dx$$

because $ds^2 = dx^2 + dy^2$

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

For trajectory $y(x)$ the travel time is $T(y(x)) = \int_{x_1}^{x_2} dx \, \frac{\sqrt{1 + (y')^2}}{v(x)} = \int_{x_1}^{x_2} dx\, L$

$$\Rightarrow \quad \frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0$$

~~But our L does not depend on~~

Get $v(x)$ from energy conservation:

$$\frac{1}{2} m \underset{\substack{\parallel \\ 0}}{v_1^2} + m g y_1 = \frac{1}{2} m v(x)^2 + m g y(x)$$

$$\Rightarrow \quad v(x) = \sqrt{2g\left(y_1 - y(x)\right)}$$

Thus $L = \frac{\sqrt{1 + (y')^2}}{\sqrt{y_1 - y}}$ for (rescaled) $\sqrt{2g}\, T = \int_{x_1}^{x_2} dx\, L$

$$\frac{\partial L}{\partial y} = \frac{\sqrt{1 + y'^2}}{+2(y_1 - y)^{3/2}} \qquad \frac{\partial L}{\partial y'} = \frac{(1 + y'^2)^{-1/2}}{2\sqrt{y_1 - y}}\, 2y' = \frac{y'}{\sqrt{1 + y'^2}\sqrt{y_1 - y}}$$

$$\Rightarrow \quad \frac{\sqrt{1 + y'^2}}{2(y_1 - y)^{3/2}} = \frac{d}{dx} \frac{y'}{\sqrt{1 + y'^2}\sqrt{y_1 - y}}$$

looks complicated...

But let us observe that $L(y, y(x))$ does not depend on x explicitly.

Then do a trick: $y'\left(\frac{\partial L}{\partial y} - \frac{d}{dx}\frac{\partial L}{\partial y'}\right) = 0$

(multiply EL equ by $y' = \frac{dy}{dx}$)

$$\Downarrow$$

$$\frac{\partial L}{\partial y} y' - y' \frac{d}{dx}\frac{\partial L}{\partial y'} = 0$$

Consider now $\frac{dL}{dx} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial L}{\partial y'}\frac{\partial y'}{\partial x} + \underbrace{\frac{\partial L}{\partial x}}_{\phi} =$

$$= \frac{\partial L}{\partial y} y' + y'' \frac{\partial L}{\partial y'}$$

$$\Rightarrow y' \frac{\partial L}{\partial y} = \frac{dL}{dx} - y'' \frac{\partial L}{\partial y'}$$

$$\Rightarrow \frac{dL}{dx} - y'' \frac{\partial L}{\partial y'} - y' \frac{d}{dx}\frac{\partial L}{\partial y'} = 0$$

$$\Rightarrow \frac{d}{dx}\left(L - y' \frac{\partial L}{\partial y'}\right) = 0$$

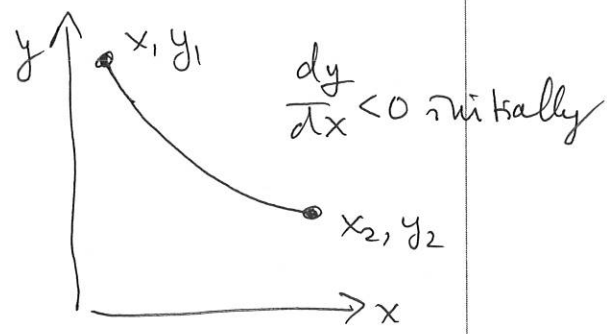
That is, the object $H(y, y(x)) = L - y' \frac{\partial L}{\partial y'}$ is <u>constant of motion!</u>

$$\frac{d}{dx} H = 0.$$

Now, $H = \frac{\sqrt{1+y'^2}}{\sqrt{y_1 - y}} - \frac{y'^2}{\sqrt{1+y'^2}\sqrt{y_1-y}} = \text{constant} = C$

$$\Rightarrow \frac{1}{\sqrt{1+y'^2}\sqrt{y_1-y}} = C \Rightarrow (1+y'^2)(y_1 - y) = \frac{1}{C^2}$$

$$\Rightarrow 1 + y'^2 = \frac{1}{C^2(y_1-y)} \equiv \frac{2a}{y_1 - y}$$



$$1 + y'^2 = \frac{2a}{y_1 - y}$$

$$y'^2 = \frac{2a - y_1 + y}{y_1 - y}$$

$$\frac{dy}{dx} = y' = -\sqrt{\frac{2a - y_1 + y}{y_1 - y}}$$

(the $\ominus$ sign is dictated by physics)

$$-dx = \sqrt{\frac{y_1 - y}{2a - y_1 + y}}\, dy$$

let $y_1 - y = 2a \sin^2\left(\frac{\varphi}{2}\right) \Rightarrow 2a - (y_1 - y) = 2a \cos^2 \frac{\varphi}{2}$

$$-dx = \sqrt{\frac{2a \sin^2 \frac{\varphi}{2}}{2a \cos^2 \frac{\varphi}{2}}}\,(-2a)\, 2 \sin\frac{\varphi}{2} \cos\frac{\varphi}{2}\, \frac{d\varphi}{2} =$$

$$= -2a \sin^2 \frac{\varphi}{2}\, d\varphi \quad \text{but } \sin^2\frac{\varphi}{2} = \frac{-1}{2}(\cos\varphi - 1)$$

$$dx = -2a \frac{1}{2}(\cos\varphi - 1)\, d\varphi = a(1 - \cos\varphi)\, d\varphi$$

$$\begin{cases} x - x_1 = a(\varphi - \sin\varphi) \\ y_1 - y = 2a \frac{1}{2}(1 - \cos\varphi) = a(1 - \cos\varphi) \\ y - y_1 = -a(1 - \cos\varphi) \end{cases}$$

$$\begin{cases} y(\varphi) = y_1 - a(1 - \cos\varphi) \\ x(\varphi) = x_1 + a(\varphi - \sin\varphi) \end{cases} \qquad \begin{cases} y(\varphi = 0) = y_1 \\ x(\varphi = 0) = x_1 \end{cases}$$

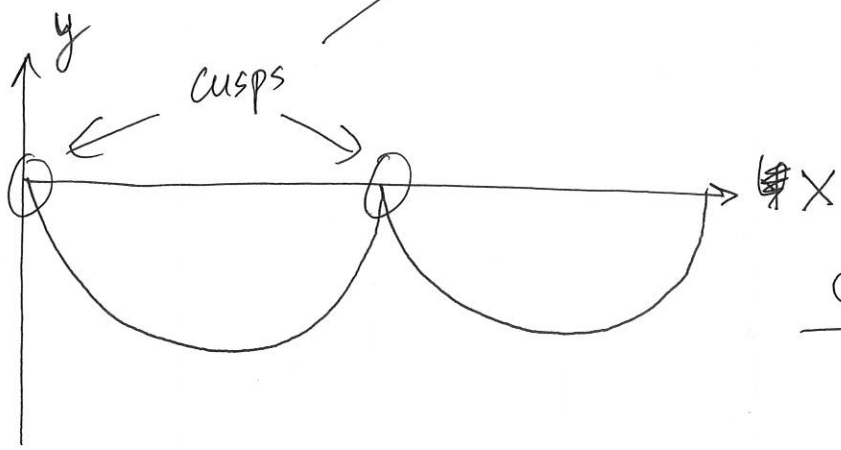
let $y(0) = x(0) = 0$

Then for $\varphi \to 0$: $y = -a\left(1 - [1 - \frac{1}{2}\varphi^2]\right) = -\frac{a}{2}\varphi^2$

$$x = a\left(\varphi - [\varphi - \frac{1}{6}\varphi^3]\right) = \frac{a}{6}\varphi^3$$

$$\Rightarrow y \sim -x^{2/3}$$

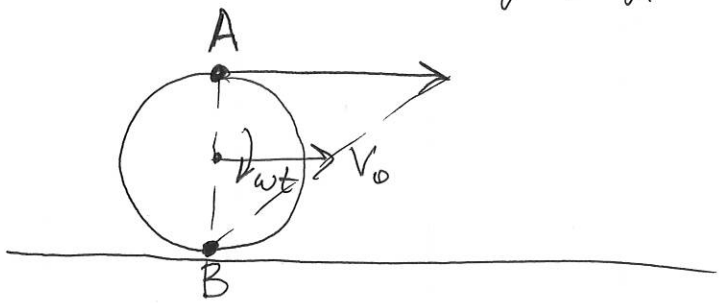
$$\frac{dy}{dx} \sim \frac{1}{x^{1/3}} \xrightarrow[x \to 0]{} \infty$$



cycloid

let $\varphi = \omega t \Rightarrow$ $x(t) - x_1 = a(\omega t - \sin\omega t)$

$$y(t) - y_1 = -a(1 - \cos\omega t)$$



$$x_A = V_0 t + R\sin\omega t$$

$$y_A = R(1 - \cos\omega t)$$

No slipping $\Rightarrow V_B = 0 \Rightarrow V_0 - R\omega = 0$

$$x = x_1 + a(\varphi - \sin\varphi)$$

$$y = y_1 - a(1-\cos\varphi) \quad , \quad y_1 - y = a(1-\cos\varphi)$$

$$dx = a(d\varphi - \cos\varphi\, d\varphi) = a(1-\cos\varphi)\, d\varphi$$

$$dy = +a(-\sin\varphi)\, d\varphi$$

$$\frac{dy}{dx} = \frac{-a\sin\varphi\, d\varphi}{a(1-\cos\varphi)\, d\varphi} = -\frac{\sin\varphi}{1-\cos\varphi}$$

$$\cancel{y_1 - y = a(1-\cos\varphi)}$$

$$\sqrt{2g}\, T = \int_{x_1}^{x_2} dx \frac{\sqrt{1+y'^2}}{\sqrt{y_1 - y}} = \int_{\varphi_1}^{\varphi_2} d\varphi\, a(1-\cos\varphi) \frac{\sqrt{1+\frac{\sin^2\varphi}{(1-\cos\varphi)^2}}}{\sqrt{a(1-\cos\varphi)}}$$

$$= \sqrt{a} \int d\varphi \sqrt{1-\cos\varphi} \frac{\sqrt{1-2\cos\varphi+\cos^2\varphi+\sin^2\varphi}}{1-\cos\varphi}$$

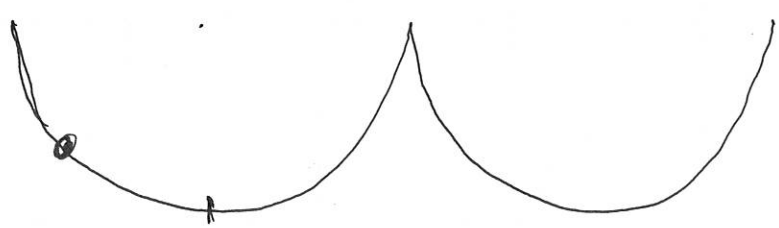
$$= \sqrt{2a} \int_{\varphi_1}^{\varphi_2} d\varphi \frac{\sqrt{1-\cos\varphi}\sqrt{1-\cos\varphi}}{1-\cos\varphi} = \sqrt{2a}\, \varphi \Big|_{\varphi_1}^{\varphi_2}.$$

The lowest point of the curve: $y(\varphi=\pi) = y_1 - a(1-(-1)) =$

$$= y_1 - 2a.$$

$$x(\varphi=\pi) = x_1 + \pi a.$$

Initial point of the curve $\varphi = 0$: $x = x_1, y = y_1$.



Constraints

$$f(x,y,z) = x^2 + 2y^2 + 3z^2 + 2xy + 2xz$$

constraint: $x^2 + y^2 + z^2 = 1$ } ~~HW01 problem~~

Generally: maximize $f(x,y)$, subject to condition $g(x,y) = \text{const}$:

$$df = f_x\,dx + f_y\,dy = 0$$

$$dg = g_x\,dx + g_y\,dy = 0 \Rightarrow dx = -\frac{g_y}{g_x}\,dy$$ (x and y are not independent!)

$$f_x\left(-\frac{g_y}{g_x}\,dy\right) + f_y\,dy = 0$$

$$\Rightarrow \frac{f_x}{g_x} = \frac{f_y}{g_y}$$ ⇐ this is ratio we denote λ:

$$\frac{f_x}{g_x} = \lambda = \frac{f_y}{g_y} \Rightarrow \begin{cases} f_x - \lambda g_x = 0 \\ f_y - \lambda g_y = 0 \end{cases}$$

Consider now $\varphi(x,y,\lambda) = f(x,y) - \lambda\, g(x,y)$

$$d\varphi = f_x\,dx + f_y\,dy - \lambda\left(g_x\,dx + g_y\,dy\right) - d\lambda\, g(x,y) = 0$$

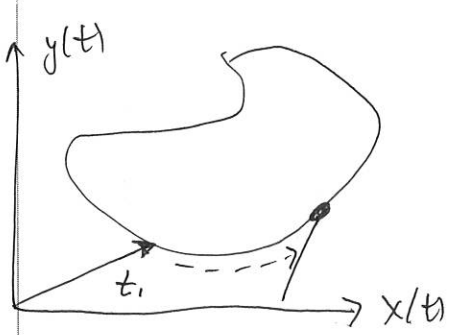
Since x, y, λ are independent variables $\Rightarrow$

$$\left.\begin{aligned} (f_x - \lambda g_x)\,dx &= 0 \\ (f_y - \lambda g_y)\,dy &= 0 \\ g(x,y)\,d\lambda &= 0 \end{aligned}\right\} \quad \begin{cases} f_x - \lambda g_x = 0 \\ f_y - \lambda g_y = 0 \\ g(x,y) = 0 \end{cases}$$

which are the same equations!

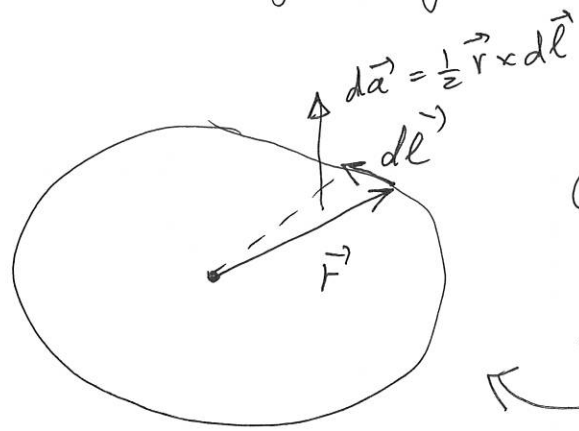
Constraints!

Ex Closed plane curve that encloses the maximum area for a given perimeter:

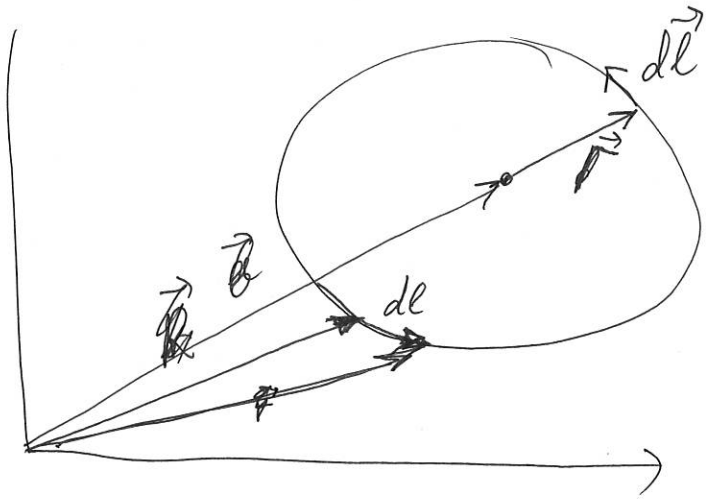


$$\text{Area } A = \frac{1}{2}\int_{t_1}^{t_2} (x\dot{y} - y\dot{x})\,dt$$

generally, $A = \frac{1}{2}\int \vec{r}\times d\vec{l}$



Geometric proof: slice the area into infinitesimal triangles.



$$A = \frac{1}{2}\oint \vec{r}\times d\vec{l} = \frac{1}{2}\oint (\vec{R} - \vec{b})\times d\vec{l}$$

where $\vec{b}$ = constant vector

But $\vec{b}\times\oint d\vec{l} = 0$ for any closed loop as the displacement $\oint d\vec{l} = 0$.

$\Rightarrow A = \frac{1}{2}\oint \vec{r}\times d\vec{l}$ for any choice of coordinates $\vec{r}$.

More formally, rewrite

$$A = \frac{1}{2}\int (x\dot{y} - y\dot{x})\,dt = \int \vec{c}\cdot d\vec{l}$$

where $d\vec{l} = dx\,\hat{x} + dy\,\hat{y}$ (displacement vector)

$$= (\dot{x}\,\hat{x} + \dot{y}\,\hat{y})\,dt$$

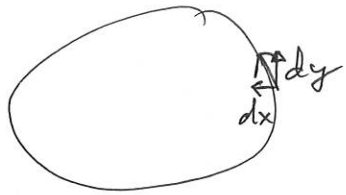
Then $\vec{c} = \frac{1}{2}(-y\,\hat{x} + x\,\hat{y})$

<u>Stokes th</u> $\oint \vec{c}\cdot d\vec{l} = \int (\vec{\nabla}\times\vec{c})\cdot d\vec{S}$

$$\vec{\nabla}\times\vec{c} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ -\frac{1}{2}y & \frac{1}{2}x & 0 \end{vmatrix} = \hat{z}\left(\partial_x\left(\tfrac{1}{2}x\right) - \partial_y\left(-\tfrac{1}{2}y\right)\right) = \hat{z}$$

$$\Rightarrow A = \oint \vec{c}\cdot d\vec{l} = \int \hat{z}\cdot d\vec{S} = \int dS = \text{AREA}$$

Perimeter $L = \int \sqrt{dx^2 + dy^2} = \int \sqrt{\dot{x}^2 + \dot{y}^2}\, dt$



$$dx = \frac{dx}{dt}\, dt = \dot{x}\, dt$$
$$dy = \dot{y}\, dt$$

$\Rightarrow$ maximize $J = \int \left(\frac{1}{2}(x\dot{y} - y\dot{x}) + \lambda \sqrt{\dot{x}^2 + \dot{y}^2} \right) dt$

$$- \lambda L \left(\frac{1}{2\pi} \right)$$

$$\frac{\partial h}{\partial x} - \frac{d}{dt} \frac{\partial h}{\partial \dot{x}} = 0 \Rightarrow \frac{1}{2}\dot{y} - \frac{d}{dt}\left(-\frac{1}{2} y + \lambda \frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) = 0$$

$$\frac{\partial h}{\partial y} - \frac{d}{dt} \frac{\partial h}{\partial \dot{y}} = 0 \Rightarrow -\frac{1}{2}\dot{x} - \frac{d}{dt}\left(\frac{1}{2} x + \lambda \frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) = 0$$

$$\Rightarrow \begin{cases} -x - \dfrac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = \text{const}, & x + \dfrac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c_1 \\ y - \dfrac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = \text{const}, & y - \dfrac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c_2 \end{cases}$$

$$\begin{cases} x - c_1 = \dfrac{-\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \\ y - c_2 = \dfrac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \end{cases} \Rightarrow (x - c_1)^2 + (y - c_2)^2 = \lambda^2$$

which is equation for a circle of radius λ.

$$L = \int \sqrt{\dot{x}^2 + \dot{y}^2}\, dt = \int \sqrt{dx^2 + dy^2}$$

~~y/A~~ let now $\left.\begin{array}{l} x = c_1 + \lambda \cos\varphi \\ y = c_2 + \lambda \sin\varphi \end{array}\right\} \begin{array}{l} (x-c_1)^2 = \cos^2\varphi\,\lambda^2 \\ (y-c_2)^2 = \sin^2\varphi\,\lambda^2 \end{array}$

$$\begin{cases} dx = -\lambda \sin\varphi\, d\varphi \\ dy = \lambda \cos\varphi\, d\varphi \end{cases}$$

$$\sqrt{dx^2 + dy^2} = \sqrt{(\lambda \sin\varphi)^2 + (\lambda\cos\varphi)^2}\, d\varphi = \lambda\, d\varphi$$

$$\Rightarrow L = \int_0^{2\pi} \lambda\, d\varphi = 2\pi\lambda$$

$$\Rightarrow (x-c_1)^2 + (y-c_2)^2 = \left(\frac{L}{2\pi}\right)^2 = \left(\frac{2\pi R}{2\pi}\right)^2 = R^2.$$

circle of radius $R = \frac{L}{2\pi}$. Area $= \pi R^2$

$$= \pi\left(\frac{L}{2\pi}\right)^2$$

$$= \frac{L^2}{4\pi}$$

Is this max area or min?

Consider square $L = 4a$ ← perimeter

$$\Rightarrow \text{area} = a^2 = \left(\frac{L}{4}\right)^2 = \frac{L^2}{16}$$

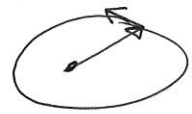
Clearly $A_{circle} = \frac{L^2}{4\pi} > A_{square} = \frac{L^2}{4 \cdot 4}$

$\Rightarrow$ circle is the figure that maximizes area.

Continuous symmetry and conserved charge. Noether's Theorem.

Consider 2d motion in external potential $U(r)$

$$L = T - U = \frac{1}{2} m\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - U(r)$$



$\dot{r}$ = radial velocity

$r\dot{\varphi}$ = angular velocity

$$\frac{\partial L}{\partial \varphi} = 0 \Rightarrow \frac{d}{dt}\frac{\partial L}{\partial \dot{\varphi}} = \frac{d}{dt}\left(mr^2\dot{\varphi}\right) = 0$$

Momentum $p_\varphi = mr^2\dot{\varphi}$ is conserved. = angular momentum

In terminology of field theory, p_φ is conserved "charge."

Noether's Th: Consider one-parameter family of transformations

$$q_\sigma \to \tilde{q}_\sigma(q, \xi)$$

here ξ = continuous parameter. Choose $\xi = 0$ to be identity transformation: $\tilde{q}_\sigma(q, 0) = q$.

Suppose next that L is invariant (does not change) under replacement $q_\sigma \to \tilde{q}_\sigma$.

Then $$0 = \frac{d}{d\xi} L(\tilde{q}, \dot{\tilde{q}}, t)\Big|_{\xi=0} = \frac{\partial L}{\partial q}\frac{\partial \tilde{q}}{\partial \xi}\Big|_{\xi=0} +$$

$$+ \frac{\partial L}{\partial \dot{q}}\frac{\partial \dot{\tilde{q}}}{\partial \xi}\Big|_{\xi=0} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)\frac{\partial \tilde{q}}{\partial \xi}\Big|_{\xi=0} +$$

$$+ \frac{\partial L}{\partial \dot{q}}\frac{\partial \dot{\tilde{q}}}{\partial \xi}\Big|_{\xi=0} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\frac{\partial \tilde{q}}{\partial \xi}\right)\Big|_{\xi=0}$$

Therefore $\Lambda = \frac{\partial L}{\partial \dot{q}}\frac{\partial \tilde{q}}{\partial \xi}\Big|_{\xi=0}$ is the conserved "charge".

For $L = \frac{1}{2} m\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - U(r)$ we can choose

the transformation as $\tilde{r}(\xi) = r$, $\tilde{\varphi}(\xi) = \varphi + \xi$.

$$\Rightarrow \quad \Lambda = \sum_\sigma \frac{\partial L}{\partial \dot{q}_\sigma} \frac{\partial \tilde{q}_\sigma}{\partial \xi}\bigg|_{\xi=0} = \frac{\partial L}{\partial \dot{r}} \underbrace{\frac{\partial \tilde{r}}{\partial \xi}\bigg|_{\xi=0}}_{=0} +$$

$$+ \frac{\partial L}{\partial \dot{\varphi}} \underbrace{\frac{\partial \tilde{\varphi}}{\partial \xi}\bigg|_{\xi=0}}_{=1} = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi} = \text{Angular momentum.}$$

Conservation of linear and angular momentum:

1) Suppose L is invariant under uniform translation of all particles in the $\hat{n}$ direction.

Then $\vec{\tilde{X}}_a = \vec{X}_a + \xi\hat{n}$

$$\Rightarrow \Lambda = \sum_a \frac{\partial L}{\partial \dot{\vec{X}}_a} \frac{\partial \vec{\tilde{X}}}{\partial \xi}\bigg|_{\xi=0} = \sum_a \underbrace{\frac{\partial L}{\partial \dot{\vec{X}}_a}}_{\vec{p}_a} \cdot \hat{n} = \hat{n} \cdot \vec{P}$$

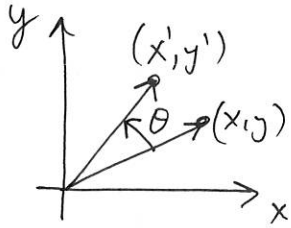
$\Rightarrow$ total momentum along direction $\hat{n}$

$$\vec{P} \cdot \hat{n} = \left(\sum_a \vec{p}_a\right) \cdot \hat{n} \quad \text{is conserved.}$$

2) Let now L be invariant under rotations around axis $\hat{n}$

$$\Rightarrow \quad \vec{\tilde{X}}_a = \vec{X}_a + \xi\,\hat{n} \times \vec{X}_a$$

For example: rotation about $\hat{z}$ $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$



For $\theta \to 0$: $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & -\theta \\ \theta & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x - \theta y \\ y + \theta x \end{pmatrix}$

$$\xi \hat{z} \times \vec{r} = \xi \hat{z} \times (x\hat{x} + y\hat{y} + z\hat{z}) = (x\hat{y} - y\hat{x})\xi$$

$$\Rightarrow \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix} = \begin{pmatrix} x - \xi y \\ y + \xi x \\ z \end{pmatrix} = \begin{pmatrix} 1 & -\xi & 0 \\ \xi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Therefore parameter ξ is just the angle of rotation θ, $\xi = \theta$:

$\Rightarrow$ under rotation indeed $\vec{\tilde{x}}_a = \vec{x}_a + \xi \hat{n} \times \vec{x}_a$

$$\Rightarrow \Lambda = \sum_a \frac{\partial L}{\partial \dot{\vec{x}}_a} \cdot \frac{\partial \vec{\tilde{x}}_a}{\partial \xi}\bigg|_{\xi=0} = \sum_a \left(\frac{\partial L}{\partial \dot{x}_a}\right) \cdot \hat{n} \times \vec{x}_a$$

$$= \sum_a \vec{p}_a \cdot \hat{n} \times \vec{x}_a = \hat{n} \cdot \sum_a \vec{x}_a \times \vec{p}_a = \hat{n} \cdot \vec{L}$$

$\Rightarrow$ total angular momentum $\vec{L}$ along axis $\hat{n}$ is conserved!

Consider spiral-like potential: $U(\rho, \varphi, z) = U(\rho, a\varphi + z)$

$\uparrow$ cylindrical coordinates

$$L = \frac{1}{2} m \left(\dot{\rho}^2 + \rho^2 \dot{\varphi}^2 + \dot{z}^2\right) - U(\rho, a\varphi + z)$$

Helical symmetry: $\tilde{\rho}(\xi) = \rho$

$\tilde{\varphi}(\xi) = \varphi + \xi$

$\tilde{z}(\xi) = z - a\xi$

leaves $U(\rho, a\varphi + z) = U(\tilde{\rho}, a(\varphi + \xi) + z - a\xi)$

$= U(\tilde{\rho}, a\tilde{\varphi} + \tilde{z})$ invariant.

$$\Rightarrow \Lambda = \frac{\partial L}{\partial \dot{\rho}} \underbrace{\frac{\partial \tilde{\rho}}{\partial \xi}\Big|_{\xi=0}}_{=0} + \frac{\partial L}{\partial \dot{\varphi}} \underbrace{\frac{\partial \tilde{\varphi}}{\partial \xi}\Big|_{\xi=0}}_{=1} + \frac{\partial L}{\partial \dot{z}} \underbrace{\frac{\partial \tilde{z}}{\partial \xi}\Big|_{\xi=0}}_{=-a} =$$

$$= m\rho^2 \dot{\varphi} + m\dot{z}(-a) = m\rho^2 \dot{\varphi} - ma\dot{z}$$

is conserved.

It is a combination of angular momentum along $\hat{z}$ and linear momentum $m\dot{z}$, multiplied by a.